

Human Resource Management System (HRMS)

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1. Project Overview

The Human Resource Management System (HRMS) is a console-based application developed using Core Java. The system helps organizations manage employee information, attendance records, leave requests, and reports efficiently. It provides a centralized platform for HR administrators to perform employee-related operations while maintaining data persistence through file handling.

The project follows Object-Oriented Programming (OOP) principles and includes features such as employee management, attendance tracking, leave management, report generation, authentication, and data storage.

2. Objectives

- To manage employee information efficiently.
- To maintain attendance records of employees.
- To handle employee leave requests.
- To generate reports for HR administrators.
- To provide secure login authentication.
- To store and retrieve data using file handling.

3. Technologies Used

Programming Language

- Java

Concepts Used

- Object-Oriented Programming (OOP)
- Classes and Objects
- Constructors
- Encapsulation
- Interfaces
- File Handling

- Exception Handling
- Collections Framework (ArrayList)

Development Environment

- Visual Studio Code (VS Code)
- JDK (Java Development Kit)

4. System Modules

4.1 Employee Management

This module allows HR administrators to:

- Add Employee
- View Employees
- Update Employee Information
- Delete Employee Records
- Search Employees

Employee details include:

- Employee ID
- Name
- Designation
- Department
- Contact Information

4.2 Attendance Management

This module allows administrators to:

- Mark Employee Attendance
- View Attendance Records

Attendance status includes:

- Present
- Absent

Attendance records are stored persistently in a text file.

4.3 Leave Management

This module allows administrators to:

- Add Leave Requests
- View Leave Requests
- Approve Leave Requests
- Reject Leave Requests

Leave details include:

- Leave ID
- Employee ID
- Leave Type
- Start Date
- End Date
- Leave Status

4.4 Report Generation

The system generates reports related to:

Employee Report

Displays employee details and total employee count.

Attendance Report

Displays:

- Total Present Employees
- Total Absent Employees

Leave Report

Displays:

- Pending Leaves
- Approved Leaves

- Rejected Leaves

4.5 Authentication Module

The authentication module restricts access to authorized HR administrators.

Features:

- Username Verification
- Password Verification
- Login Validation

Only valid users can access the system.

5. Data Persistence

The project stores data using text files.

Files Used

employees.txt

Stores employee information.

attendance.txt

Stores attendance records.

leaves.txt

Stores leave request information.

This ensures data remains available even after the application is closed and restarted.

6. Project Structure

Employee Module

- Employee.java
- EmployeeService.java

Attendance Module

- Attendance.java
- AttendanceService.java

Leave Module

- LeaveRequest.java
- LeaveService.java

Authentication Module

- AuthService.java

Reporting Module

- ReportGenerator.java
- ReportService.java

Main Module

- Main.java

7. Features Implemented

Employee Features

- Add Employee
- View Employee
- Update Employee
- Delete Employee
- Search by Employee ID
- Search by Employee Name
- Search by Department

Attendance Features

- Mark Attendance
- View Attendance

Leave Features

- Add Leave Request
- Approve Leave Request

- Reject Leave Request
- View Leave Requests

Reports

- Employee Report
- Attendance Report
- Leave Report

Security

- Login Authentication

Storage

- File-Based Data Persistence

8. Exception Handling

The system handles:

- Invalid Employee IDs
- Duplicate Employee IDs
- File Handling Errors
- Invalid User Inputs
- Missing Records

Appropriate error messages are displayed to the user.

9. Advantages of the System

- Easy to use
- Reduces manual HR work
- Maintains employee records efficiently
- Provides quick employee search
- Stores data permanently
- Generates useful reports
- Supports attendance and leave management

10. Limitations

- Console-based interface
- Single-user access
- Uses text files instead of a database
- Static administrator credentials

11. Future Enhancements

- Web-based HRMS
- Database Integration (MySQL)
- Employee Self-Service Portal
- Email Notifications
- Role-Based Access Control
- Advanced Analytics Dashboard
- Cloud Deployment

12. Conclusion

The Human Resource Management System (HRMS) successfully fulfills the requirements of employee management, attendance tracking, leave management, report generation, authentication, and data persistence. The project demonstrates the practical implementation of Core Java concepts, Object-Oriented Programming principles, file handling, and exception handling while providing an efficient solution for managing HR-related activities.